

ENERGY CHANGES – BOND ENERGY CALCULATIONS

KEY VOCABULARY

Bond energy: the energy needed to break one mole of a bond.

Breaking bonds: endothermic — energy is taken in.

Making bonds: exothermic — energy is released.

ΔH : energy to break - energy released making bonds.

COMMON MISCONCEPTIONS

~~— Breaking bonds releases energy.~~

✓ Breaking bonds takes IN energy (endothermic).

~~— Making bonds takes in energy.~~

✓ Making bonds RELEASES energy (exothermic).

~~— A negative ΔH is endothermic.~~

✓ A negative ΔH is exothermic.

ENERGY CHANGE

$$\Delta H = (\text{bonds broken}) - (\text{bonds made})$$

CONVERSION REMINDER

- value = exothermic • + value = endothermic

1 BREAK OR MAKE

State whether breaking bonds is exo- or endothermic and why. [2 marks]

2 CALCULATE ΔH

Bonds broken = 1500 kJ; bonds made = 1800 kJ. Calculate ΔH . [2 marks]

Answer: _____ kJ/mol

3 CLASSIFY

A reaction has $\Delta H = -120$ kJ/mol. State whether it is exo- or endothermic. [1 mark]

4 WORK BACKWARDS

$\Delta H = +60$ kJ; bonds made = 940 kJ. Calculate the energy to break the bonds. [2 marks]

Answer: _____ kJ

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Energy is taken _____ to break bonds and given _____ when bonds are made.

broken

made

in

out

6 EXPLAIN SIGN

Explain why an exothermic reaction has a negative ΔH . [2 marks]
